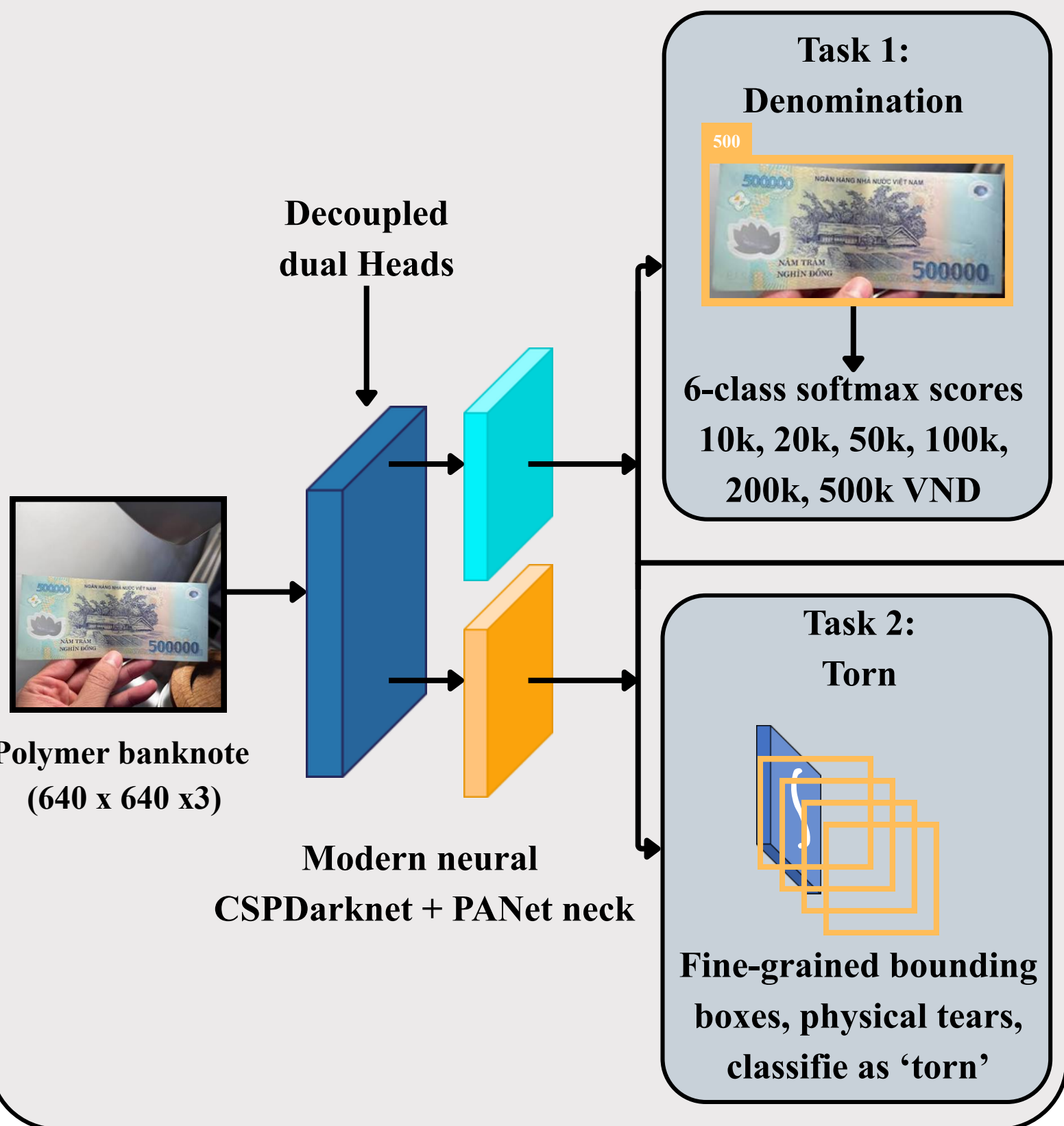
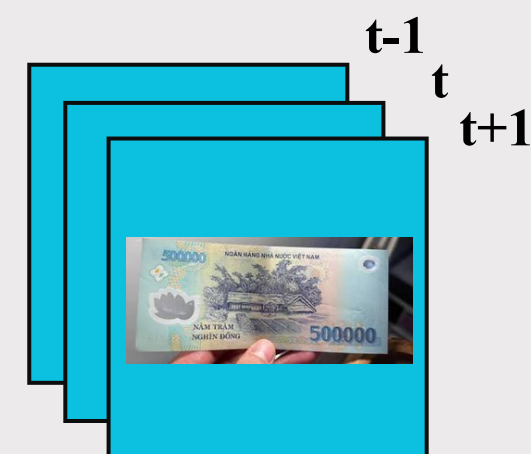


Joint YOLOv8n Detector



Multi-Frame Consensus Aggregation



Multi-Frame Buffer

Confidence-Weighted Denomination Voting

$$c^* = \arg \max_{c \in \mathcal{C}_{\text{denom}}} \sum_{t \in \mathcal{T}_{\text{verify}}} s_c^{(t)}$$

Two-Frame Defect Confirmation Rule

$$\sum_{t \in \mathcal{T}_{\text{verify}}} \mathbb{I}(\text{torn detected}) \geq 2$$

